

Section I

10 marks

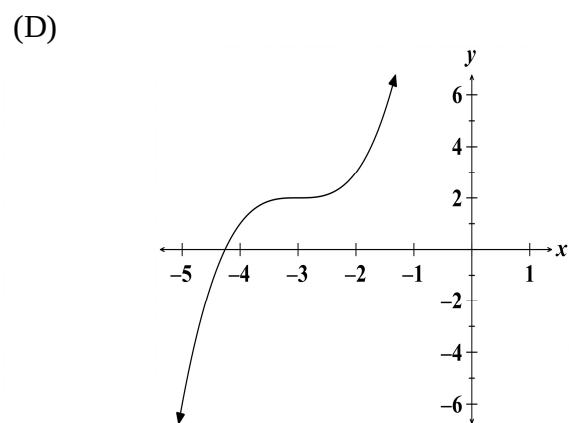
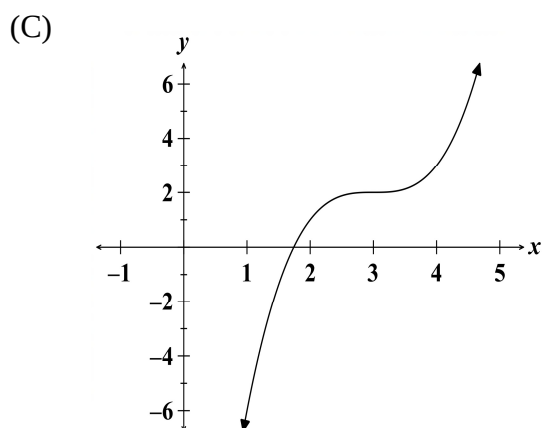
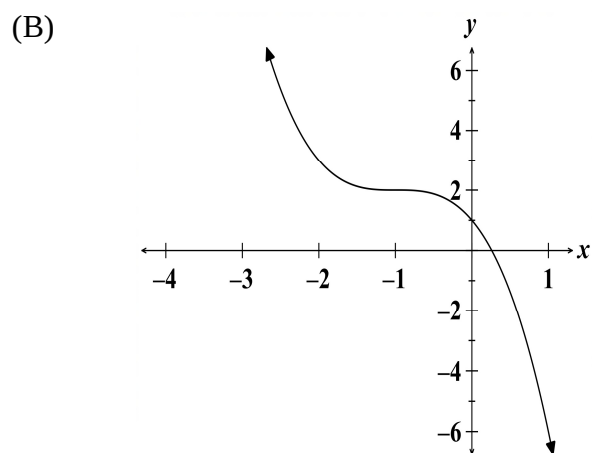
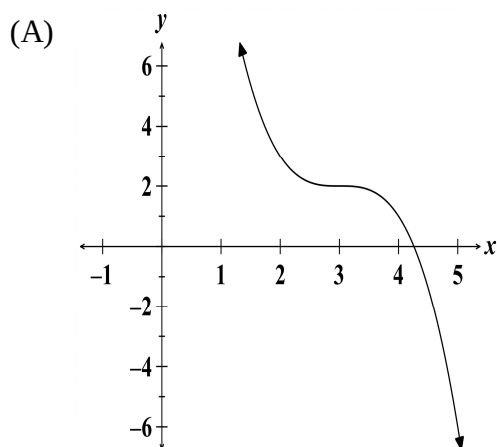
Attempt Questions 1-10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet provided for Questions 1-20

- 1 The function with rule $f(x) = (x-1)^2$ has a range of $[1, \infty)$.
The domain of f could be
- (A) $x \in (-\infty, 0]$
(B) $x \in (-\infty, 0)$
(C) $x \in (2, \infty)$
(D) $x \in (-\infty, 0) \cup (2, \infty)$
- 2 Given that $\theta = 42^\circ 30'$, which of the following is the value of θ in radians?
- (A) 43.5π
(B) 0.7418
(C) 2435.07
(D) 1.3642
- 3 What is the solution to the equation $\log_3(2x-5) = 2$?
- (A) $3\frac{1}{2}$
(B) $6\frac{1}{2}$
(C) $5\frac{1}{2}$
(D) 7

- 4 Which diagram best shows the graph of the curve $y = 2 - (x - 3)^3$?



- 5 Anika answered five multiple choice questions. Each question offered four possible answers. If Anika guessed every answer, what is the probability that she guessed all five questions correctly?

(A) $5 \times \left(\frac{1}{4}\right)$

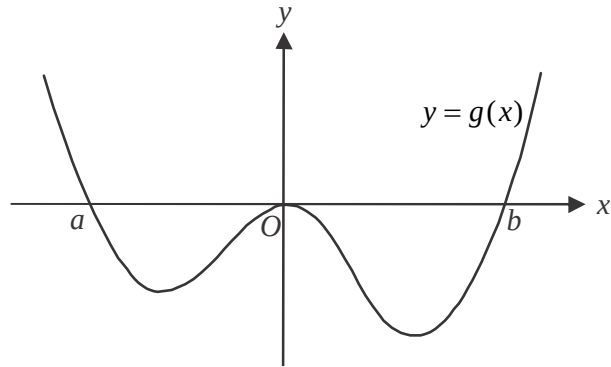
(B) $\frac{1}{5} \times \frac{1}{4} \times \frac{1}{3} \times \frac{1}{2} \times 1$

(C) $\left(\frac{1}{4}\right)^5$

(D) $\left(\frac{1}{5}\right)^4$

6

The graph of the function g is shown below.



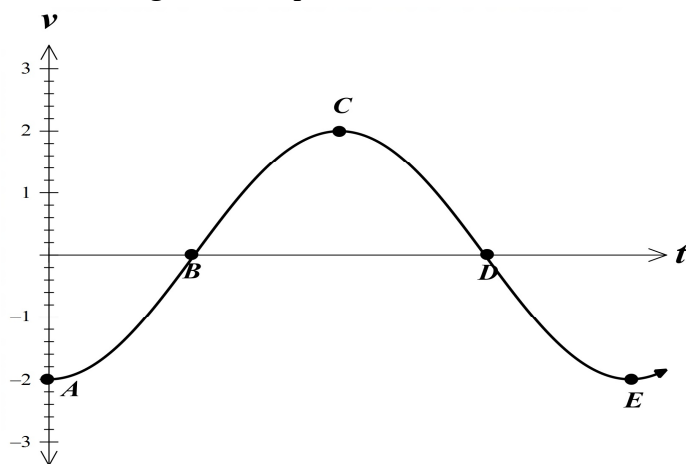
The rule for g could be:

- (A) $g(x) = -x(x+a)(x+b)$
- (B) $g(x) = -x^2(x-a)(x-b)$
- (C) $g(x) = x(x-a)(x-b)$
- (D) $g(x) = x^2(x-a)(x-b)$

7 Which is the derivative of $\frac{\sin x}{\cos x + 1}$?

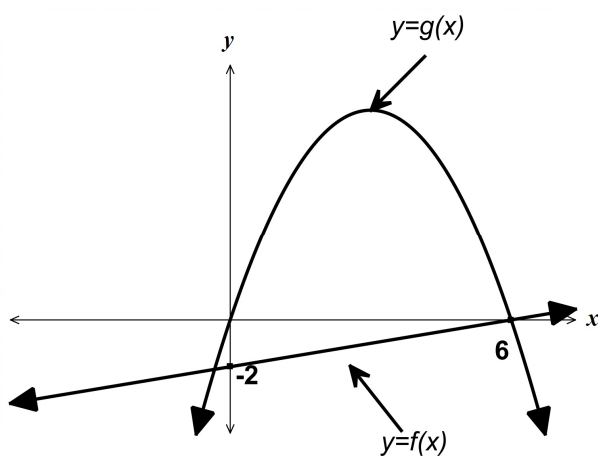
- (A) $\frac{\sin x + \cos x}{(\cos x + 1)^2}$
- (B) $\frac{2}{(\cos x + 1)^2}$
- (C) $\frac{\cos^2 x + \cos x - \sin^2 x}{(\cos x + 1)^2}$
- (D) $\frac{1}{\cos x + 1}$

- 8 A particle is moving in a straight line. The graph below shows the **velocity** (v) of the particle for time $t \geq 0$. If the particle is initially at rest at the origin, at which point does the particle have the greatest displacement?



- (A) D
(B) B
(C) C
(D) E

9



If $\int_0^6 [g(x) + f(x)] dx = 20$, find $\int_0^6 g(x) dx$

- (A) 32
(B) 8
(C) 14
(D) 26

10

The geometric progression $1 + \frac{1}{2-x} + \frac{1}{(2-x)^2} + \dots$ has a limiting sum for which values of x ?

- (A) $1 < x < 3$
- (B) $x < 1$ or $x > 3$
- (C) $x \neq 2$
- (D) $0 < x < 2$

Section II

90 marks

Attempt Questions 11-16

Allow about 2 hours and 45 minutes for this section

Answer each question in the appropriate writing booklet. Extra writing booklets are available.

In Questions 11-16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use a separate Writing Booklet

- (a) Rationalise the denominator of $\frac{4}{3-\sqrt{7}}$. 2
- (b) If $y = \sqrt{x^2 + 1}$, find $\frac{dy}{dx}$. 2
- (c) Evaluate $\lim_{p \rightarrow 3} \left(\frac{p^3 - 27}{p^2 - 9} \right)$. 2
- (d) Find the equation of the tangent to the curve $y = e^{x^2}$ at the point where $x = 1$. 2
- (e) Find $\int \frac{1}{(x+5)^3} dx$. 2
- (f) Find the co-ordinates of the centre and radius of the circle $x^2 - 6x + y^2 = 7$. 2
- (g) Find $\int_{\frac{3\pi}{8}}^{\frac{\pi}{2}} \sec^2 \left(\frac{2x}{3} \right) dx$. 3

End of Question 11

Question 12 (15 marks) Use a separate Writing Booklet.

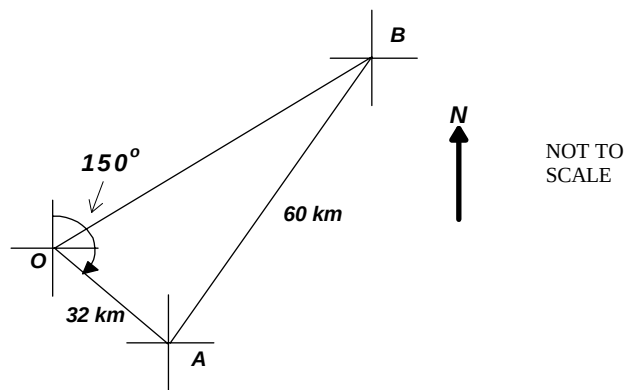
- (a) Find the natural domain of $y = \frac{1}{x^2 - 4}$. Write your answer using set interval notation. 1

- (b) Find, in general form, the equation of the line with y-intercept -2 and angle of inclination 60° . 2

- (c) Explain how to shift the graph of $y = x^2$ to obtain the function $y = (x + 4)^2 + 7$ 2

- (d) Given that $f(x) = 5x - 2$ and $g(x) = x^2 + 3$, find $f(g(a))$ and $g(f(a))$. 2

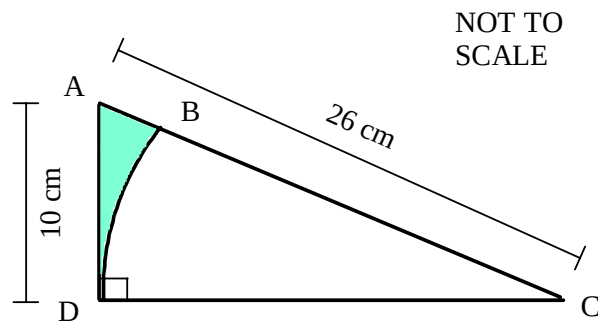
(e)



Marcel travelled 32 km from O to A at a bearing of 150° T. He then travelled 60 km to B at a bearing of 20° T.

- (i) Copy the diagram in your solution booklet and show why $\angle OAB = 50^\circ$. 1
- (ii) Use the Cosine Rule to find the length of OB to 2 decimal places. 2
- (iii) Find the bearing of O from B to the nearest minute. 2

(f)



In the above diagram, $\angle ADC = 90^\circ$, BD is an arc of a circle centre C . $AD=10$ cm and $AC=26$ cm.

- (i) Find $\angle ACD$ in radians. 1
- (ii) Hence or otherwise, find the magnitude of the shaded area bounded by ABD . 2

End of Question 12

Question 13 (15 marks) Use a separate Writing Booklet.

- (a) A sculpture along a motorway in Europe is made up of many wooden poles placed side by side in a straight line in ascending order. The difference in heights between any two adjacent poles is constant. The shortest pole is 0.42m and the tallest measures 4.35m. The sum of heights of all the poles is 76.32m.

(i) Find the number of poles used. 2

(ii) Find the difference in heights between any two adjacent poles. 2

- (b) (i) Show that $\frac{x+2}{5x^2+7x-6} = \frac{1}{5x-3}$. 1

(ii) Hence find a , given that $\int_1^a \left(\frac{x+2}{5x^2+7x-6} \right) dx = \frac{1}{5} \ln 6$. 3

- (c) Fayed's local library is open for five hours on Saturdays. The number (N) of people in the library during a particular Saturday, is given by:

$$N = -t^4 + 4t^3 + 125 \quad 0 \leq t \leq 5$$

Where time (t) is measured in hours.

(i) Find the maximum number of people in the library during this five-hour period. 3

(ii) Find the time at which the number of people in the library was increasing most rapidly. 2

(iii) Sketch the curve $N = -t^4 + 4t^3 + 125 \quad 0 \leq t \leq 5$ showing all critical points. 2

End of Question 13

Question 14 (15 marks) Use a separate Writing Booklet.

- (a) The number of deliveries made to a business on any given work day can be represented by the random variable X .

The probability distribution of X is shown in the table below.

x	0	1	2	3
$\Pr(X = x)$	0.3	0.4	0.2	0.1

The mean number of deliveries is 1.1.

- (i) Find the variance of X . 2
- (ii) Find the probability that on a work day when there is a delivery, the number of deliveries is two or more. 2
- (b) Solve the equation $\log_e(x) + \log_e(3x + 2) = 2\log_e(x + 1)$ for x , where $x > 0$. 3

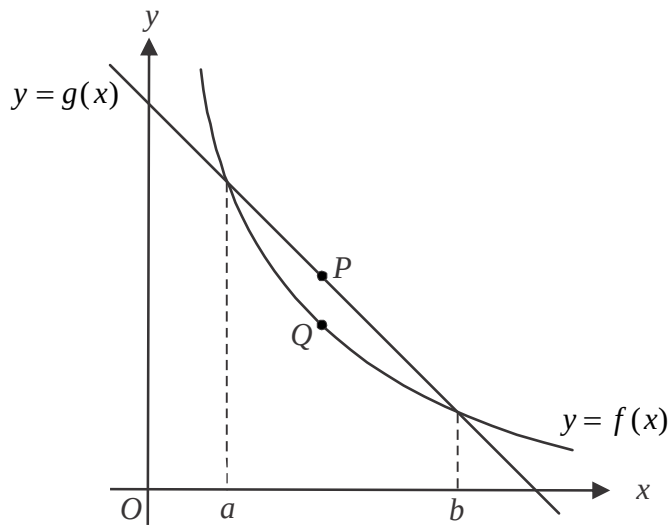
A continuous random variable X has a probability density function given by

(c)

$$f(x) = \begin{cases} 2\sin(4x), & \text{in domain } \left[0, \frac{\pi}{4}\right]. \\ 0, & \text{otherwise} \end{cases}$$

- (i) Find $\Pr\left(X < \frac{\pi}{6}\right)$ 2
- (ii) Find $\Pr\left(X < \frac{\pi}{8} \mid X < \frac{\pi}{6}\right)$ 3

- (d) Let $f(x) = \frac{4}{x}$ with domain $(0, \infty)$ and $g(x) = 5 - x$ with domain all x .
The graphs of f and g intersect at the points where $x = a$ and $x = b$ as shown below.



Point P lies on the graph of g between $x = a$ and $x = b$.
Point Q lies on the graph of f between $x = a$ and $x = b$.
Point P lies vertically above point Q .

- | | | |
|------|--|----------|
| (i) | Find an expression in terms of x for L , the length of the line segment PQ . | 1 |
| (ii) | Find the maximum length of the line segment PQ . | 2 |

End of Question 14

Question 15 (15 marks) Use a separate Writing Booklet.

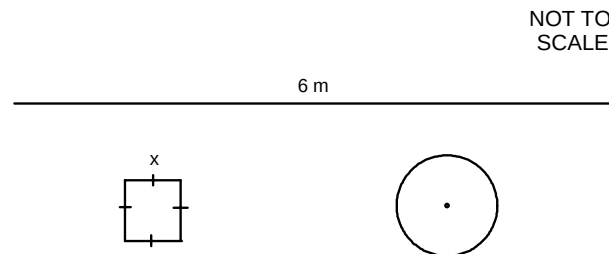
(a) Solve $2^{2x} - 3 \times 2^{x+1} + 5 = 0$. 2

(b) A curve is defined as $y = x \ln x$ for $x \geq 1$.

(i) Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$. 2

(ii) By considering part (i), indicate the gradient and concavity of $y = x \ln x$ and hence sketch the graph of $y = x \ln x$ for $x \geq 1$. 3

(c)



A piece of string of length 6 metres is cut into two pieces. One piece forms a square with sides $x \text{ cm}$ and the other piece forms a circle.

(i) Show that the radius (r) of the circle in terms of x is given by 2

$$r = \frac{3 - 2x}{\pi}.$$

(ii) Hence find the lengths of the two pieces of string which obtain the minimum area. 3

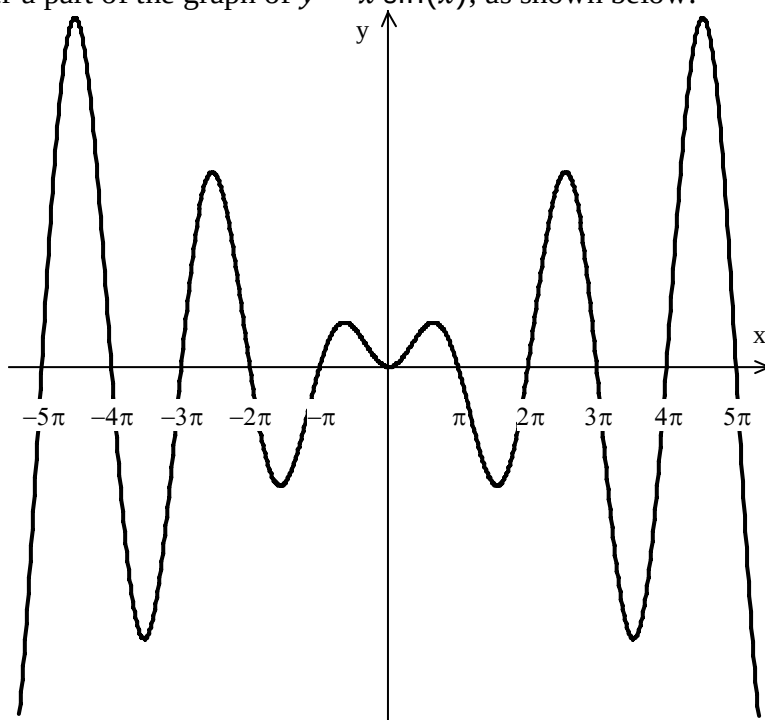
(d) Find the value of k so that the following curves $y = \frac{x^4}{k}$ and 3

$y = -3x^3 - 2x^2$ intersect at more than two points.

End of Question 15

Question 16 (15 marks) Use a separate Writing Booklet.

- (a) Consider a part of the graph of $y = x \sin(x)$, as shown below:



- (i) Given that:

2

$$\int (x \sin(x)) dx = \sin(x) - x \cos(x) + C,$$

Evaluate:

$$\int_{n\pi}^{(n+1)\pi} (x \sin(x)) dx,$$

when n is a positive even integer or 0. Give your answer in simplest form.

- (ii) Given that:

1

$$\int (x \sin(x)) dx = \sin(x) - x \cos(x) + C,$$

Evaluate:

$$\int_{n\pi}^{(n+1)\pi} (x \sin(x)) dx,$$

when n is a positive odd integer. Give your answer in simplest form.

- (b) Declan is playing a game by continually rolling a normal single six-sided die. He wins the game if he rolls even numbers on consecutive rolls of the die.

- (i) Find the probability that the game ends before the seventh roll.

3

- (ii) Find the probability that Declan will end the game with an odd number of rolls.

2

- (c) A particle is moving in a straight line. Initially, it is travelling to the left at 1 cm/sec. Its acceleration as a function of time (t) is given by
- $$a = 2 \cos 2t + 2 \sin 2t \text{ for } 0 \leq t \leq \pi$$
- where time and displacement are measured in seconds and cm respectively.

- (i) Show that the velocity of the particle is given by $v = \sin 2t - \cos 2t$. **1**
- (ii) Find when the particle changes direction. **2**
- (iii) Find when the particle has maximum velocity. **2**
- (iv) Find the total distance travelled in the first $\frac{\pi}{2}$ seconds. *Leave your answer in simplified exact form.* **2**

End of Paper